L4

12/09

Definition:- Suppose (A, B) is a matrix pair. If $\tilde{A}_{22}$ in a controllability form is Hurwitz, then we say the system is stabilizable.

$$\boxed{\dot{\tilde{x}}_2 = \tilde{A}_{22}\tilde{x}_2}$$

- Suppose $\tilde{A}_{22}$ is not Hurwitz, then $\exists$ an initial condition $\tilde{x}_2(0)$ such that $x(t)$ does not go to zero as $t \to \infty$.

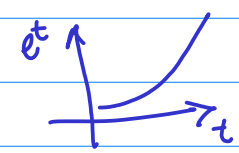
e.g. $\tilde{A}_{22} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

$\tilde{x}_2(t) = e^{\tilde{A}_{22}t}\tilde{x}_2(0)$

$= \begin{bmatrix} e^{t} & 0 \\ 0 & e^{-t} \end{bmatrix}\tilde{x}_2(0)$

$\tilde{x}_2(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$\tilde{x}_2(t) = \begin{bmatrix} e^{t} \\ 0 \end{bmatrix}$



- Conversely, if $\tilde{A}_{22}$ is Hurwitz, then for every initial condition $x(0)$, $\exists$ i/p s.t. $x(t) \to 0$ as $t \to 0$.

$$A = \left[\begin{array}{c|cc} \overbrace{1}^{\tilde{A}_{11}} & \overbrace{0 \quad 0}^{\tilde{A}_{12}} \\ \hline 0 & -1 & 0 \\ 0 & 0 & -1 \end{array}\right] \qquad B = \left[\begin{array}{c} 1 \\ \hline 0 \\ 0 \end{array}\right]$$

$\tilde{A}_{22}$

$$\begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} = \begin{bmatrix} e^{t} & 0 & 0 \\ 0 & e^{t} & 0 \\ 0 & 0 & e^{-t} \end{bmatrix}\begin{bmatrix} x_{10} \\ x_{20} \\ x_{30} \end{bmatrix} + \int_0^t e^{A(t-\tau)} \underbrace{B}_{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}} u(\tau)\, d\tau$$

$$\begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} = \begin{bmatrix} \boxed{e^{t}x_{10} + \int_0^t e^{(t-\tau)} u(\tau)\, d\tau} \\ e^{t}x_{20} \\ e^{-t}x_{30} \end{bmatrix}$$

$\to 0$ as $t \to \infty$

$t \to \infty$

$x_{10} = 0$

$x_{20} = 1$

$x_{30} = 0$

$$\begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} = \begin{bmatrix} \int_0^t e^{t-\tau}\, d\tau \\ e^{t} \\ 0 \end{bmatrix}$$

$\lim_{t \to \infty} x(t) \neq 0$

Theorem: The pair (A, B) is controllable if and only if for each $\lambda \in \mathbb{C}$ the following holds

$$\boxed{\text{rank}\begin{bmatrix} A - \lambda I & B \end{bmatrix} = n}$$

PBH

Popov-Belevitch-Hautus Test

Proof: $\text{rank}\begin{bmatrix} A-\lambda I & B \end{bmatrix} = n$ $\quad \underline{\underline{\lambda \in \mathbb{C}}}$ $\quad \lambda \in \sigma(A)$

$a \Rightarrow b$

not $a \Leftarrow$ not b

Only if:- ✓ (A,B) controllable $\Rightarrow$ $\text{rank}\begin{bmatrix} \lambda I - A & B \end{bmatrix} = n \quad \forall \lambda \in \mathbb{C} \; (\lambda \in \sigma(A))$

Contrapositive: ✓ (A,B) uncontrollable $\Leftarrow$ $\text{rank}\begin{bmatrix} \underline{\underline{\lambda I - A}} & B \end{bmatrix} < n$ for some $\lambda \in \mathbb{C}$

$\text{rank}\begin{bmatrix} \lambda I - A & B \end{bmatrix} < n$ $\qquad n \times (n+m)$

$\Rightarrow \exists\, v$ such that $\quad v^* \begin{bmatrix} \lambda I - A & B \end{bmatrix} = 0$

$$\Rightarrow \begin{bmatrix} \lambda v^* - v^* A & v^* B \end{bmatrix} = 0$$

(i) $\lambda v^* - v^* A = 0 \Rightarrow v^* A = \lambda v^*$

(ii) $v^* B = 0$

$$v^* A = \lambda v^* \Rightarrow v^* A^2 = \lambda v^* A \Rightarrow v^* A^2 = \lambda^2 A$$

$$\vdots$$

$$v^* A^k = \lambda^k A$$

$$v^* \begin{bmatrix} B & AB & \cdots & A^{n-1}B \end{bmatrix} = 0$$

$\therefore v \in \ker \mathcal{C}_{AB}^*$

$\text{rank}\, \mathcal{C}_{AB} < n \Rightarrow$ Uncontrollability

$v^* B = 0$

$$\boxed{\begin{aligned} &v^* A = \lambda v^* \\ &v^* AB = \lambda v^* B = 0 \end{aligned}}$$

$v^* A^2 = \lambda^2 v^*$

$v^* A^2 B = \lambda^2 v^* B = 0$

If: $\text{rank}\begin{bmatrix} \lambda I - A & B \end{bmatrix} = n \Rightarrow (A,B)$ controllable

Contrapositive: $\text{rank}\begin{bmatrix} \lambda I - A & B \end{bmatrix} < n \Leftarrow (A,B)$ uncontrollable

Given: $\exists\, T \in \mathbb{R}^{n \times n}$ s.t. $\quad T^{-1} A T = \underbrace{\begin{bmatrix} \tilde{A}_{11} & \tilde{A}_{12} \\ 0 & \tilde{A}_{22} \end{bmatrix}}_{\tilde{A}} \qquad T^{-1} B = \underbrace{\begin{bmatrix} \tilde{B}_1 \\ 0 \end{bmatrix}}_{\tilde{B}}$

Let $\lambda \in \sigma(\tilde{A}_{22})$

$\text{rank}(ABC) \leq \min\left(\text{rank}(A), \text{rank}\,B, \text{rank}\,C\right)$

$$\text{rank}\,[A-\lambda I \quad B] = \text{rank}\,[T\tilde{A}T^{-1} - \lambda T T^{-1} \quad T\tilde{B}]$$

$$= \text{rank}\left(T\,[\tilde{A}-\lambda I \quad \tilde{B}] \begin{bmatrix} T^{-1} & 0 \\ 0 & I_{m\times m} \end{bmatrix} \right)$$

$$= \text{rank}\left([\tilde{A}-\lambda I \quad \tilde{B}] \right)$$

$$= \text{rank}\left(\begin{bmatrix} \tilde{A}_{11}-\lambda I_{r\times r} & \tilde{A}_{12} & \tilde{B} \\ 0 & \tilde{A}_{22}-\lambda I_{(n-r)} & 0 \end{bmatrix} \right) \neq n$$

$\lambda \in \sigma(\tilde{A}_{22})$

$\text{rank}\,(\tilde{A}_{22}-\lambda I_{n-r}) < n-r$

PBH test

λ

$\mathbb{C}^+$

Corollary: (A, B) is stabilizable if and only if $\text{rank}\,[A-\lambda I \quad B] = n$ holds for $\lambda \in \mathbb{C}^+$